Faithful Spatial Attribution via Total-Variation Input Masking, with Application to El Niño Forecasting

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Abstract

Attribution methods are widely used to explain the predictions of convolutional neural networks (CNNs) over spatial fields, but spatial dependence makes attribution ill-posed: when neighbouring inputs are highly correlated, predictive credit can be assigned almost arbitrarily among them, and post-hoc saliency maps become noisy and unstable. We study an in-model alternative in which a learnable input mask is trained jointly with the predictor and regularized by a *structured-sparsity* penalty that combines an ℓ_1/ℓ_2-style sparsity term with a total-variation (TV) term encouraging spatial smoothness—the fused-lasso / Markov-random-field prior familiar from spatial statistics, applied here to a learned attribution map.

On synthetic data with known ground-truth signal and distractor regions, we show that (i) the TV penalty significantly improves the faithfulness of learned attribution and, applied as a modular add-on, improves not only our mask but also stochastic-gate baselines (L_0 and STG); and (ii) our TV+sparsity mask occupies a favourable point on the faithfulness–accuracy frontier, delivering faithful attribution without degrading—and slightly improving—predictive performance, whereas sparsity-only gates achieve marginally sharper localization only by collapsing the predictor. We apply the method to El Niño prediction from sea-surface temperature, comparing the learned masks against a fused-lasso statistical reference, and show that coarse tiled masks retain interpretability at a fraction of the training cost. We are explicit about limitations, including a failure mode in which sparsity-driven masks are captured by cheap spurious shortcuts.

Keywords: Interpretable machine learning; spatial attribution; structured sparsity; fused lasso; convolutional neural networks; sea surface temperature.

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1 Introduction

Deep learning has become central to spatiotemporal environmental prediction—sea surface temperature (SST), atmospheric circulation, precipitation—where convolutional neural networks (CNNs) model high-dimensional, nonlinear fields with strong predictive skill (Rasp et al. 2020, Bi et al. 2023, Pathak et al. 2022, Ham et al. 2019). In high-stakes geoscientific applications, however, predictions are only actionable if we can say *where* on the field a model is drawing its information, and interpretability has become a research priority in its own right (Toms et al. 2020, Mamalakis et al. 2022). A central obstacle is that environmental inputs are strongly spatially dependent: neighbouring grid cells are highly correlated, so predictive credit can be distributed almost arbitrarily among them. As a result, gradient-based and perturbation-based attribution maps are typically diffuse, noisy, and unstable, and need not reflect the structure a domain scientist would call physical.

Two broad families of attribution exist. *Post-hoc* methods explain an already-trained network: gradient saliency (Simonyan et al. 2013), integrated gradients (Sundararajan et al. 2017), Grad-CAM (Selvaraju et al. 2017), and SHAP (Lundberg & Lee 2017) are widely used but can be unstable or manipulable and, being external to the model, need not reflect its internal feature use (Adebayo et al. 2018, Ghorbani et al. 2019). *In-model* methods instead build attribution into training: gradient-regularized explanations (Ross et al. 2017, Ross & Doshi-Velez 2018), class activation mapping (Zhou et al. 2016), learned attention (Hu et al. 2018, Woo et al. 2018), and—closest to our setting—learnable feature gates that select inputs jointly with the predictor, including L_0/hard-concrete gates (Louizos et al. 2018), stochastic gates (Yamada et al. 2020), and instance-wise selectors (Chen et al. 2018, Yoon et al. 2019). These learned gates penalize *sparsity* but treat input locations as exchangeable, with no notion of spatial structure—precisely what is needed when the inputs form a coherent field.

Our approach is in the in-model family: a learnable input mask, trained jointly with the predictor, that gates the input before prediction. Its distinguishing ingredient is a *total-variation* (TV) penalty on the mask in addition to sparsity, so that the learned attribution is encouraged to be both compact and spatially contiguous. This pairing is exactly the structured-sparsity prior of the fused lasso (Tibshirani et al. 2005) and total-variation denoising (Rudin et al. 1992) from spatial statistics, here imposed on a learned attribution map rather than on regression coefficients. The idea of combining TV with sparse attention has appeared inside per-token attention transforms for image captioning (Martins et al. 2021), and TV-regularized masks have been optimized *post hoc*, one per image, against a frozen classifier (Fong & Vedaldi 2017, Dabkowski & Gal 2017); our contribution is to bring the smoothness prior to a single, globally-learned, in-model gate, to show on ground-truth data that it measurably improves faithfulness across gating mechanisms, and to apply it to environmental forecasting. We complement prior model-based interpretability such as prototype reasoning (Chen et al. 2019) and concept bottlenecks (Koh et al. 2020).

Dropout, a widely used regularization technique in deep learning, operates by randomly masking neural units during training to prevent overfitting (Hinton et al. 2012, Srivastava et al. 2014). In CNNs, variants such as spatial dropout (Tompson et al. 2015, Wu & Gu 2015), Cutout (DeVries & Taylor 2017), and random erasing (Zhong et al. 2020) have been proposed to mask contiguous spatial regions and encourage robustness to occlusion.

Beyond random masking, partial convolution (Liu et al. 2018) and gated convolution (Yu et al. 2019) offer mechanisms for handling genuinely missing data or selectively attending to informative spatial locations. Gated mechanisms have also been used extensively in generative audio and language models (van den Oord et al. 2016, Dauphin et al. 2017), and more recently in environmental prediction tasks (Zhang et al. 2024). Layer masking, proposed by Balasubramanian & Feizi (2023), applies learned binary masks at intermediate layers to prevent distribution shifts caused by masked inputs. These methods collectively offer models structured decision processes and increased resilience to random noise.

1.1 Nonlinear Spatiotemporal Modeling in Environmental Science

Traditional statistical models for environmental forecasting often rely on linear or quasi-linear dynamics, such as autoregressive models, Gaussian processes, and reduced-rank approximations (Banerjee et al. 2003, Cressie 2015). However, nonlinearity has long been acknowledged as essential for capturing the dynamics of environmental systems (Wikle & Hooten 2010). Early work from the statistics community, including quadratic and bilinear extensions of state-space models, provided important foundations for nonlinear spatiotemporal modeling (Wikle & Berliner 2001). These models predate the deep learning surge and remain central to understanding the dynamics of geophysical systems.

More recent statistical approaches, such as Zammit-Mangion & Wikle (2020) and Wikle & Zammit-Mangion (2023), integrate deep neural networks with hierarchical Bayesian frameworks to capture complex nonlinearities while preserving uncertainty quantification.

On the deep learning front, CNNs and ConvLSTMs (Shi et al. 2015) have been used to forecast meteorological and oceanographic variables, while benchmarks such as WeatherBench (Rasp et al. 2020) have catalyzed comparative research in data-driven weather prediction. Subsequent models such as Pangu-Weather (Bi et al. 2023) and FourCastNet (Pathak et al. 2022) demonstrate that purely data-driven models can approach and even surpass numerical weather prediction (NWP) models at shorter lead times. These methods are increasingly being compared or combined with statistical spatiotemporal approaches, such as those proposed by Zammit-Mangion & Wikle (2020), to evaluate tradeoffs between interpretability, uncertainty quantification, and predictive power.

2 Methods

We propose a model-integrated attribution framework for convolutional neural networks (CNNs) based on learnable input masking. The goal of our approach is to embed spatial interpretability directly into the predictive model, such that saliency maps arise as a natural product of training rather than requiring post hoc analysis. This is achieved by integrating a learnable masking module into the architecture and regularizing it to produce sparse and spatially coherent masks.

2.1 Model Architecture

Let $x \in \mathbb{R}^{H \times W}$ denote an input spatial feature map. The model learns a real-valued spatial mask $m \in [0,1]^{H \times W}$, which is applied elementwise to the input via Hadamard product, yielding a masked input $\tilde{x} = x \odot m$. This masked input is passed to a convolutional neural network $f_\theta(\cdot)$, yielding prediction $\hat{y} = f_\theta(\tilde{x})$. The mask m is itself a function of the input, computed by a gating mechanism $g_\phi(x)$, where ϕ denotes the parameters of the mask module.

To construct the mask, each spatial location k is associated with a learnable importance score $z_k^{(m)} \in \mathbb{R}$. These scores are passed through a temperature-controlled sigmoid activation:

$$m_k = \sigma\left(\frac{z_k^{(m)}}{\tau}\right),$$

where $\sigma(\cdot)$ denotes the standard sigmoid function and $\tau > 0$ is a temperature parameter controlling the sharpness of the activation. The masked input is then given by

$$\tilde{x}_k = m_k \cdot x_k,$$

where each pixel is modulated by its corresponding mask value. This mechanism allows the model to dynamically suppress or retain information from each spatial location, based on its learned relevance to the prediction task. The mask acts as a differentiable gating layer, encouraging the model to focus on a sparse and interpretable subset of input regions.

2.2 Loss Function and Regularization

Training proceeds by minimizing a composite loss function that balances predictive performance with interpretability. The loss is defined as:

$$\mathcal{L} = \mathrm{BCE}(\hat{y}, y) + \lambda_{\mathrm{sp}} \cdot \frac{1}{K} \sum_k |m_k|^\gamma + \lambda_{\mathrm{TV}} \cdot \frac{1}{|\mathcal{N}|} \sum_{(i,j)\in\mathcal{N}} |m_i - m_j|^\gamma \tag{1}$$

- $\mathrm{BCE}(\hat{y}, y)$ is the binary cross-entropy loss between the predicted logits $\hat{y}$ and the true label y.
- The sparsity penalty (weighted by λ_{sp}) encourages the mask to be sparse, limiting the number of significant spatial locations.
- The total variation term (weighted by λ_{TV}) promotes spatial smoothness in the mask by penalizing the absolute differences between neighboring pixel values. Here, $\mathcal{N}$ denotes the set of adjacent pixel pairs.

All regularization terms are normalized by the number of spatial units (or pairs) to ensure consistent behavior across different input resolutions. The exponent parameter $\gamma \in \{1, 2\}$ controls the form of the penalization. When $\gamma = 1$, the regularization promotes sparsity in the style of an ℓ_1-norm penalty, while $\gamma = 2$ corresponds to an ℓ_2-style penalty that encourages small but diffuse activations. In our experiments, both values produced qualitatively similar saliency masks, with $\gamma = 2$ typically yielding smoother and lower-magnitude mask activations. This choice offers a tunable trade-off between promoting sharp selectivity and preserving gradient stability during optimization.

2.2.0.1 A structured-sparsity prior.

The two penalties in (1) together impose a *structured-sparsity* prior on the mask: the sparsity term limits the number of active locations, while the total-variation term couples neighbouring locations so that active regions are spatially contiguous rather than scattered. This is precisely the prior underlying the fused lasso (Tibshirani et al. 2005) and total-variation denoising (Rudin et al. 1992), and is equivalent to a maximum-a-posteriori estimate under a Laplacian Markov-random-field smoothness prior combined with a sparsity-inducing marginal—a combination long used for spatial regularization in statistics. What is new here is the *target* of the prior: rather than regularizing regression coefficients or a post-hoc per-image perturbation mask (Fong & Vedaldi 2017, Dabkowski & Gal 2017), or a per-token attention transform (Martins et al. 2021), we regularize a single, globally-learned, in-model attribution gate that is trained jointly with the predictor. In Section 3 we show on ground-truth data that the smoothness term measurably improves attribution faithfulness, and that this benefit is not specific to our gate but transfers to other learned gates (L_0, STG) as a modular add-on.

2.3 Temperature Annealing

To encourage discrete, interpretable mask behavior, we apply a sigmoid activation with a temperature parameter $\tau > 0$ that is annealed over the course of training. The temperature controls the sharpness of the sigmoid response: larger values of τ produce smoother, more graded activations, while smaller values yield sharper, near-binary outputs.

As a general rule, we initialize $\tau > 1$ to allow for soft, flexible mask values during early training, and gradually reduce τ to a value below 1 to promote sharper, more selective gating. For example, we linearly anneal τ from 2.0 to 0.5 across training epochs. This schedule encourages exploration in the early stages of learning and supports convergence toward sparse, high-confidence mask values as training progresses.

Critically, this temperature annealing also helps avoid degenerate solutions where all mask values saturate toward 0 or 1 simultaneously. By maintaining intermediate τ values during training, the model preserves meaningful gradients and can differentiate between relevant and irrelevant input regions.

2.4 Training Procedure

The full model, including both the predictive network and the masking mechanism, is trained end-to-end using optimization methodology such as stochastic gradient descent with the Adam optimizer. All components are fully differentiable: the mask values m_k are smooth functions of underlying latent variables $z_k^{(m)}$, which enables gradients to flow from the loss through the masked input and back to both the mask parameters ϕ and the network parameters θ during backpropagation.

Specifically, the gradient of the loss with respect to each importance score $z_k^{(m)}$ is computed via the chain rule using the derivative of the sigmoid activation:

$$\frac{\partial \mathcal{L}}{\partial z_k^{(m)}} = \frac{\partial \mathcal{L}}{\partial m_k} \cdot \sigma' \left(\frac{z_k^{(m)}}{\tau} \right) \cdot \frac{1}{\tau},$$

where $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ is the derivative of the sigmoid function. This enables efficient gradient-based optimization without the need for numerical differentiation or reinforcement-style gradient estimators.

2.5 Computational Considerations

The addition of a learnable input mask introduces extra computational cost during training, with the extent of this overhead depending strongly on the spatial resolution of the input rather than the number of training examples. The gating mechanism requires learning a set of latent importance scores $\{z_k^{(m)}\}$, which are passed through a temperature-controlled sigmoid and applied elementwise to the input. Although this adds negligible memory or compute for small spatial domains, the cost becomes more pronounced as the spatial resolution increases.

In particular, the total variation penalty—which computes finite differences between adjacent spatial elements—scales with the number of spatial locations and must be evaluated and backpropagated at every iteration. While this operation is vectorizable and parallelizable, it becomes a computational bottleneck for high-resolution grids (e.g., 64×64 or larger). In contrast, increasing the number of training samples (batch size or dataset size) has minimal impact on this cost, as the masking operation is applied independently across batches.

To address this scalability issue, we also consider a tiled version of the mask. The input domain is divided into K non-overlapping tiles (e.g., 10×10), and each tile $k \in \{1, \ldots, K\}$ is assigned a learnable importance score $z_k^{(m)}$. These scores are transformed into mask values via a temperature-controlled sigmoid: $m_{G_k} = \sigma\left(z_k^{(m)}/\tau\right)$, and the masked input is constructed by broadcasting the tile-level gate to all pixels in the tile: $\tilde{x}_{i,j} = m_{G_k} \cdot x_{i,j}$, where G_k denotes the tile index to which location (i, j) belongs.

This tiling strategy dramatically reduces the number of learned mask parameters—from one per pixel to one per tile—and eliminates the need to compute fine-grained total variation penalties. As a result, training time and memory usage are significantly reduced for high-resolution spatial data. Although this approach produces coarser attribution maps, the loss in granularity can be offset by improved semantic clarity: in many real-world forecasting settings, highlighting important regions (e.g., ocean basins, pressure zones) may be more interpretable and actionable than identifying individual pixels.

Despite the increased training cost, our approach remains far more efficient than perturbation-based attribution techniques such as SHAP or occlusion, which require repeated forward passes for each input. Once trained, both the per-pixel and tiled mask variants incur negligible overhead at inference time, producing predictions and saliency maps in a single forward pass.

3 Simulation Study: Faithful Attribution under Known Ground Truth

Because faithfulness cannot be measured directly on real data, we evaluate it on synthetic data with a known predictive structure, following the now-standard practice of benchmarking explanations against ground truth in the geosciences (Mamalakis et al. 2022). We ask three questions: (i) does the total-variation penalty improve the faithfulness of a learned mask; (ii) is any such benefit specific to our gate, or does it transfer to other in-model gates; and (iii) how does our mask compare to both post-hoc and learned-gate baselines on faithfulness *and* predictive performance.

3.1 Data and metrics

Each image is 24×48 with i.i.d. Gaussian background noise $x_{i,j} \sim \mathcal{N}(0, 0.5^2)$, and the dataset has $N = 500$ samples. Two circular *signal* regions (A at $(10, 15)$ and B at $(14, 30)$, radius 4) and two *distractor* regions (radius 3) each receive a per-sample $\mathcal{N}(0, 1)$ coefficient. The binary label is an XOR over the mean intensities of A and B—the label is 1 iff exactly one exceeds 0.5—so the task is nonlinear (requiring a CNN) and the distractors are, by construction, uninformative. We score each method's $H \times W$ importance map against the known regions by Intersection-over-Union (IoU, using the top-10% most salient pixels) and by attribution mass (the fraction of total map mass inside a region). Higher signal overlap and lower distractor overlap are better.

3.2 Methods compared

We compare three families on a shared two-convolution backbone. *Post-hoc* attributions are computed on an ungated CNN—gradient saliency (Simonyan et al. 2013), integrated gradients (Sundararajan et al. 2017), Grad-CAM (Selvaraju et al. 2017), and gradient SHAP (Lundberg & Lee 2017), all via Captum (Kokhlikyan et al. 2020). *Learned in-model gates* are trained jointly with the predictor: an L_0 hard-concrete gate (Louizos et al. 2018), a stochastic gate (STG) (Yamada et al. 2020), and a CBAM-style spatial attention gate (Woo et al. 2018). *Our mask* is the temperature-annealed sigmoid gate of Section 2 with sparsity and TV penalties, reported both with and without the TV term (the within-method ablation). All results are averaged over 15 random seeds; comparisons use the paired Wilcoxon signed-rank test.

3.3 Total variation improves faithfulness

Table 1 and Figure 2 report faithfulness and predictive performance. Removing the TV term from our mask reduces signal IoU from 0.747 to 0.453 (paired Wilcoxon $p = 10^{-4}$): without spatial coupling the mask selects a sparse, incomplete subset of the signal, whereas the TV term fills in the contiguous predictive region. Every learned gate suppresses the distractors strongly (IoU ≤ 0.01), while post-hoc gradient maps leak substantially onto them (IoU 0.15–0.22).

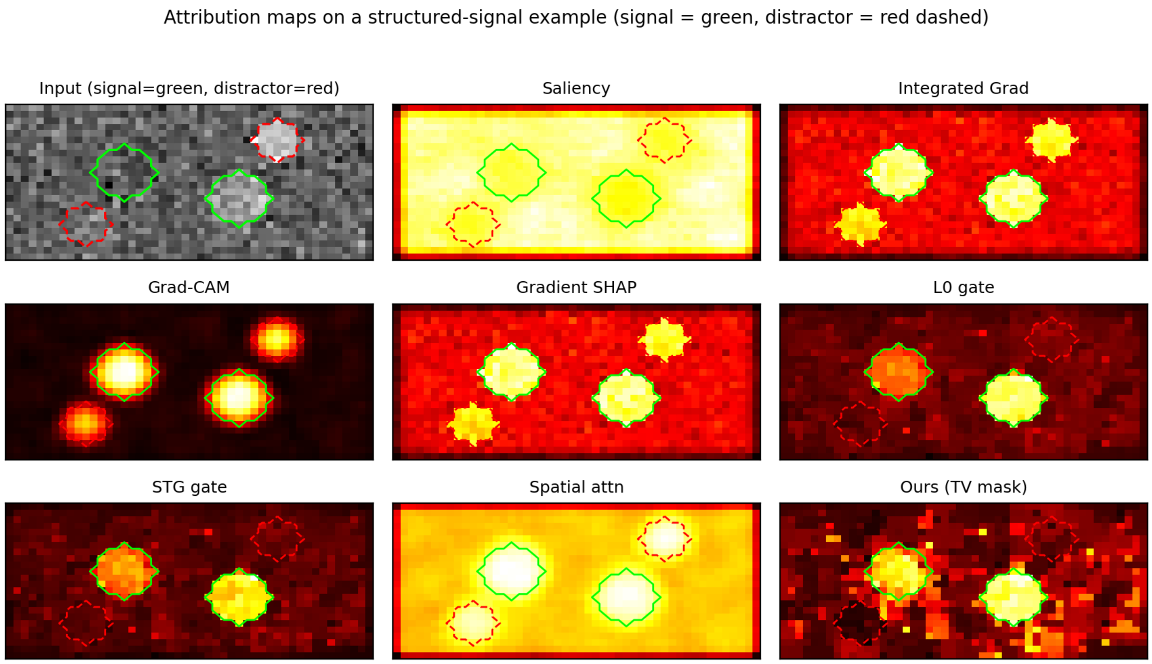

Figure 1: Attribution maps from each method on a representative test image. Signal regions are outlined in green, distractors in red dashed. Post-hoc gradient maps are diffuse and leak onto the distractors; the learned gates concentrate on the signal, and our TV mask produces a compact, contiguous attribution over the true regions.

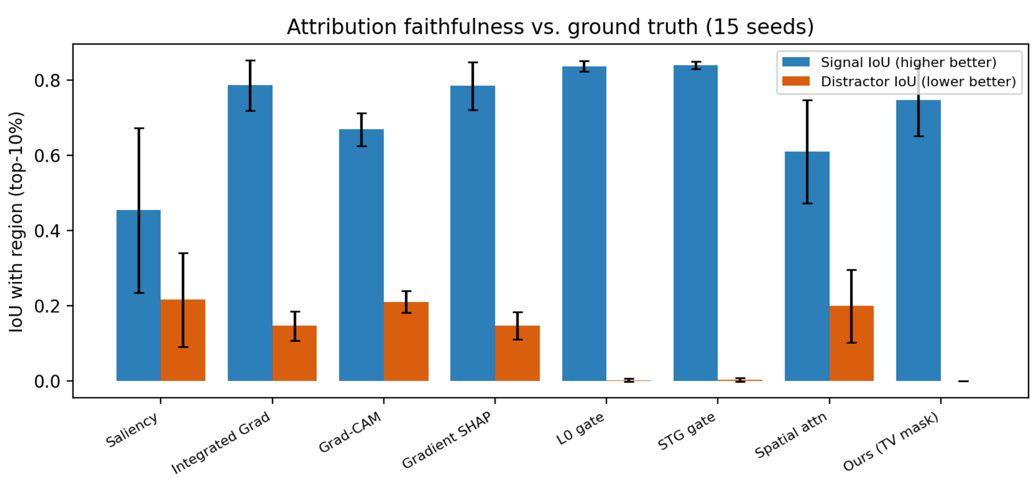

Figure 2: Signal- and distractor-region IoU per method (mean ± SD over 15 seeds). Learned gates suppress distractors; the TV term lifts our mask's signal overlap without inflating distractor overlap.

Table 1: Attribution faithfulness and predictive performance on the structured-signal simulation (mean over 15 seeds). Signal IoU: higher is better; distractor IoU: lower is better. The last column is the paired Wilcoxon p-value for signal IoU against Ours (TV).

Method	**Signal IoU**	**Distr. IoU**	**Acc**	**F1**	p **vs ours**
Saliency	0.454	0.215	0.673	0.539	10^{-4}
Integrated gradients	0.786	0.146	0.673	0.539	0.17
Grad-CAM	0.669	0.210	0.673	0.539	0.009
Gradient SHAP	0.785	0.146	0.673	0.539	0.14
L_0 gate	0.836	0.002	0.626	0.419	0.002
STG gate	0.840	0.003	0.579	0.276	0.002
Spatial attention	0.610	0.199	0.675	0.547	0.003
Ours (no TV)	0.453	0.000	0.723	0.631	10^{-4}
Ours (TV)	0.747	0.000	0.716	0.620	—

3.4 The smoothness prior transfers across gates

Because total variation regularizes any differentiable spatial map, it can be added to other in-model gates as a modular term. Sweeping the sparsity weight reveals that adding TV significantly improves the L_0 and STG gates as well, provided they are not already saturated: at moderate sparsity the gain in signal IoU is +0.07 for L_0 ($p = 0.012$) and +0.13 for STG ($p = 0.008$); at very high sparsity both gates already localize tightly and TV adds little (a ceiling effect). The smoothness prior therefore behaves as a general improvement to learned spatial attribution rather than a quirk of our particular gate. We fixed the TV weight across methods rather than tuning it per gate; a sensitivity analysis is left to the supplement.

3.5 A faithfulness–accuracy frontier

Sweeping the sparsity weight also traces out a frontier between localization and prediction (Figure 3). The sparsity-only gates (L_0, STG) reach the highest signal IoU (≈ 0.84), but only at low predictive performance: their best attainable F1 is $\approx$ 0.46–0.48, at or below the ungated baseline, because hard sparsification discards information the predictor needs. Our TV+sparsity mask instead reaches F1 ≈ 0.63— significantly higher than either gate (paired Wilcoxon $p = 0.002$) and above the ungated baseline—while still attaining signal IoU up to 0.80 (a small but significant deficit of 0.045 relative to the sparsity-only ceiling, $p = 0.043$). For an attribution mechanism this is the more useful operating point: a faithful explanation that does not require degrading the model it explains. The soft, smooth mask attenuates rather than deletes inputs, which both preserves predictive signal and—through the TV prior—keeps the attribution spatially coherent.

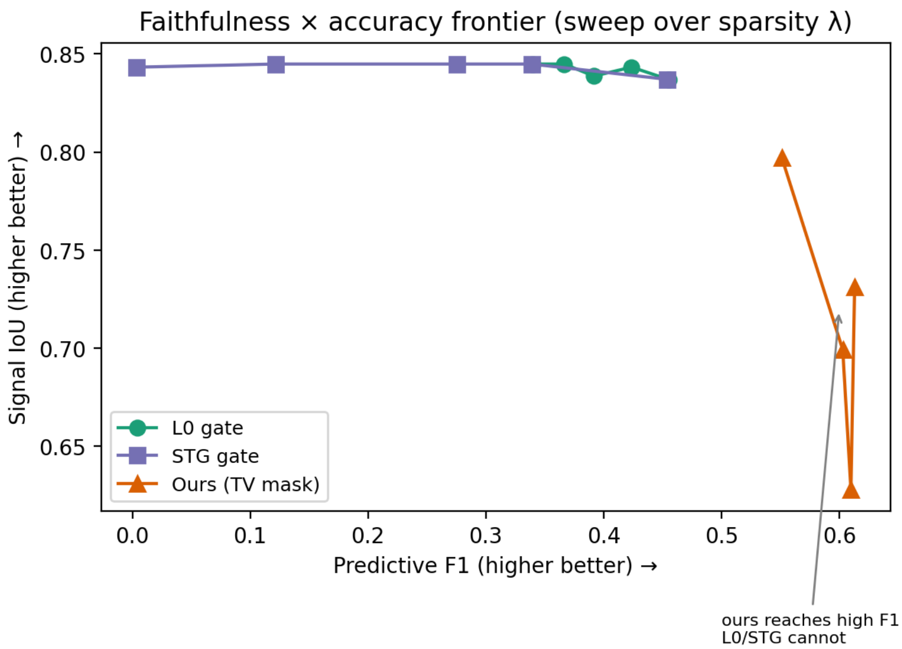

Figure 3: Faithfulness–accuracy frontier as the sparsity weight is varied (TV on). Sparsity-only gates occupy the high-IoU, low-F1 region; our TV mask occupies the high-F1 region they cannot reach, at strong signal IoU and zero distractor overlap.

3.6 Limitation: capture by spurious shortcuts

The sparsity penalty rewards concentrating mask mass on the *cheapest* predictive region. When a non-causal feature is cheaply predictive—for instance a single region whose sign is correlated with the label—the mask can be drawn onto it in preference to a more distributed causal signal. In a variant in which a spurious region encoded a noisily-flipped copy of the label, our mask concentrated on the spurious region (signal IoU 0.06) while post-hoc integrated gradients still recovered the causal regions (IoU 0.71). We therefore do *not* claim robustness to spurious correlation; characterizing when sparsity-driven masking resists versus is captured by shortcut features is important future work.

4 Application: Predicting El Niño from Sea-Surface Temperature

We now apply the masking framework to a real spatiotemporal forecasting problem with strong spatial dependence—predicting El Niño from sea-surface temperature (SST). Deep networks forecast the El Niño–Southern Oscillation with useful skill (Ham et al. 2019), and interpreting *where* on the Pacific they draw that skill is of direct scientific interest (Toms et al. 2020). SST is also an ideal stress test for the attribution-under-dependence problem: the equatorial Pacific is highly autocorrelated, so predictive credit is intrinsically hard to localize. We ask two questions: does input gating preserve predictive skill, and do the learned masks agree with a principled statistical attribution?

4.1 Data and set-up

We use monthly SST anomalies over the tropical Pacific (15°S–15°N, 170°E–100°W), deseasonalized against the monthly climatology and down-sampled to a 30×90 grid. The binary label is an El Niño event—the Niño-3.4 anomaly (5°S–5°N, 190°–240°E) exceeding 0.5°C—at a lead of k months; the event rate is $\approx 27\%$. We report leads $k = 3$ and $k = 6$. Crucially we use a *temporal holdout*: the model trains on the earlier 80% of months and is validated on the later 20%. A random split would leak across SST's strong temporal autocorrelation and inflate apparent skill; we found this inflation to be substantial (e.g. lead-3 AUROC dropped from 0.92 under a random split to 0.90 under the temporal holdout, and lead-6 from 0.76 to 0.70).

The backbone is a DeepCNN: five 3×3 convolutions ($1 \to 32 \to 64 \to 64 \to 128 \to 128$), each with batch normalization and ReLU, followed by global average pooling and an MLP head ($128 \to 64 \to 32 \to 1$, dropout 0.3). We train the ungated baseline and the pixel, 2×2, and 4×4 gates of Section 2 with sparsity ($\lambda_{\text{sp}} = 10^{-2}$) and total-variation ($\lambda_{\text{TV}} = 0.2$) penalties and temperature annealing ($\tau : 2.0 \to 0.01$), using Adam and multiplicative input dropout. The TV weight was selected by a small sweep over $\{0.01, 0.05, 0.2, 0.5\}$: $\lambda_{\text{TV}} = 0.2$ maximized mask agreement with the statistical reference (below) at unchanged AUROC, with larger values beginning to reduce skill. All numbers are means over four seeds.

4.2 Predictive performance

Table 2 reports accuracy, F1, and AUROC. At lead 3 the model has strong skill (AUROC ≈ 0.90); at lead 6 out-of-time skill is modest (AUROC ≈ 0.70)—six-month ENSO prediction from SST alone is genuinely difficult, and our temporal holdout avoids the optimism of random cross-validation. The key observation for our purposes is that *gating does not degrade prediction*: gated and ungated models are within error of one another at both leads.

Consistent with the simulations, the mask buys interpretability essentially for free on the predictive side.

Table 2: El Niño prediction under a temporal holdout (mean over 4 seeds). "Agree" is the Pearson correlation between the seed-averaged mask and the magnitude of the lagged-correlation reference (Section 4.3); higher means the mask concentrates where the statistical signal is.

Lead	**Gate**	**Accuracy**	**F1**	**AUROC**	**Agree**
3 mo	None	0.77 ± 0.03	0.66	0.866	—
3 mo	Pixel	0.80 ± 0.00	0.69	0.897	-0.02
3 mo	Tiled 2×2	0.79 ± 0.02	0.69	0.898	0.01
3 mo	Tiled 4×4	0.78 ± 0.03	0.67	0.876	0.14
6 mo	None	0.67 ± 0.08	0.47	0.635	—
6 mo	Pixel	0.65 ± 0.04	0.51	0.694	0.37
6 mo	Tiled 2×2	0.64 ± 0.08	0.51	0.699	0.47
6 mo	Tiled 4×4	0.64 ± 0.08	0.49	0.637	0.50

4.3 A statistical reference for attribution

Real SST has no ground-truth attribution, so we validate the masks by *convergent validity* against a principled statistical reference: the per-cell lagged point-biserial correlation between the SST anomaly and the future label (Figure 4, left). This map shows the expected structure—a tight equatorial warm-tongue band at lead 3 that broadens into the sub-tropics by lead 6. We also fit a fused-lasso logistic regression (ℓ_1 + TV penalties on the per-cell coefficients), the linear statistical sibling of our mask. It is, however, hampered by collinearity: with $p \gg n$ and near-indistinguishable neighbouring equatorial cells, the coefficients are unstable even under strong regularization (Figure 4, right)— itself a vivid illustration of the attribution problem. We therefore use the robust univariate correlation map as the reference.

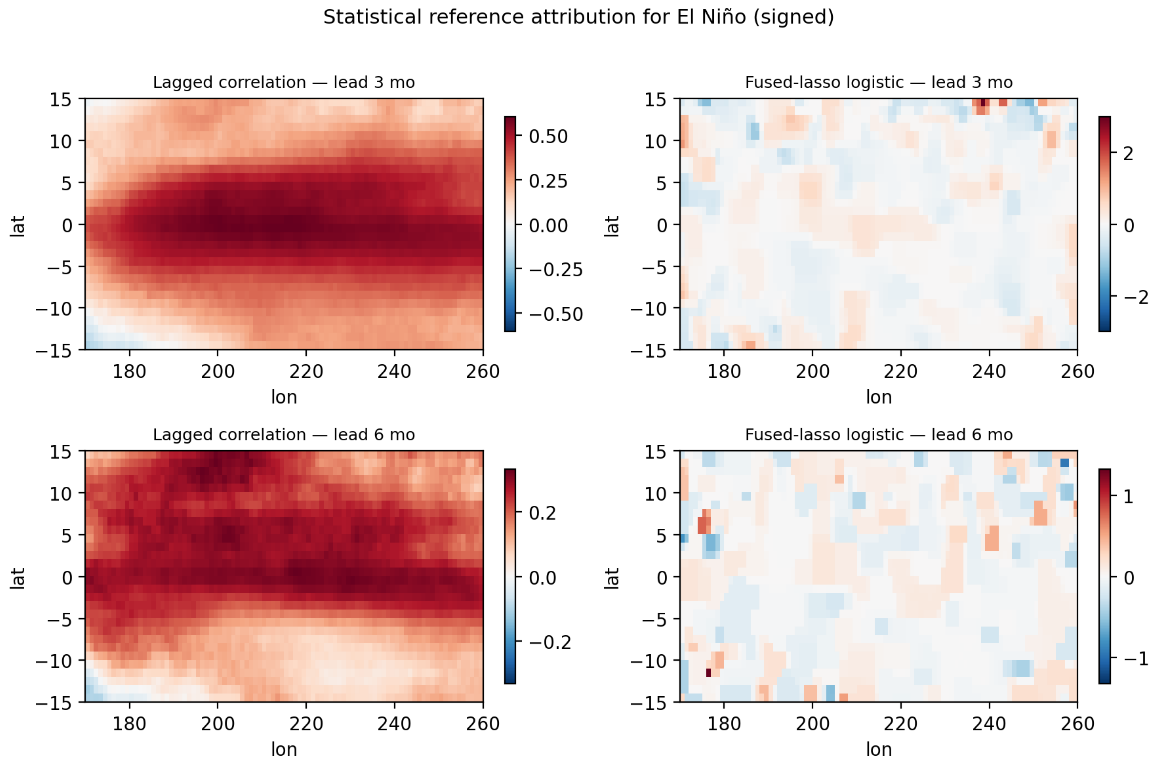

Figure 4: Statistical reference attribution for El Niño (signed). The lagged-correlation map (left) cleanly recovers the equatorial signal; the multivariate fused-lasso logistic (right) is destabilized by collinearity among the highly correlated equatorial cells.

4.4 Do the masks agree with the statistical signal?

Figure 5 places the learned masks beside the correlation reference, and the final column of Table 2 quantifies agreement as the Pearson correlation between the mask and the magnitude of the reference. The result splits sharply by lead. At **lead 6** the masks agree moderately (Pearson 0.37–0.50), and the *tiled* masks agree better than the pixel mask (0.47–0.50 vs. 0.37): coarse gating is both cheaper and cleaner. At **lead 3**, agreement is essentially zero, and this is robust across all TV weights we tried.

This contrast is not a defect of the method but a demonstration of its premise. At lead 3 the El Niño signal is so strong and persistent across the equatorial Pacific that the event can be predicted from almost any sub-region; with many individually-sufficient predictors, the input attribution is genuinely ill-posed, and no penalty pins the mask onto the "true" band. At lead 6 the signal is weaker and more spatially localized, so the mask is forced to concentrate and aligns with the statistical reference. In other words, spatial dependence makes attribution ambiguous exactly when the signal is most redundant—the motivation of this work, now visible in real data. Accordingly we rest the interpretability claims of this paper on the controlled simulation of Section 3, where ground truth exists, and treat the El Niño masks as a real-world illustration of both the promise (lead 6) and the intrinsic limits (lead 3) of spatial attribution.

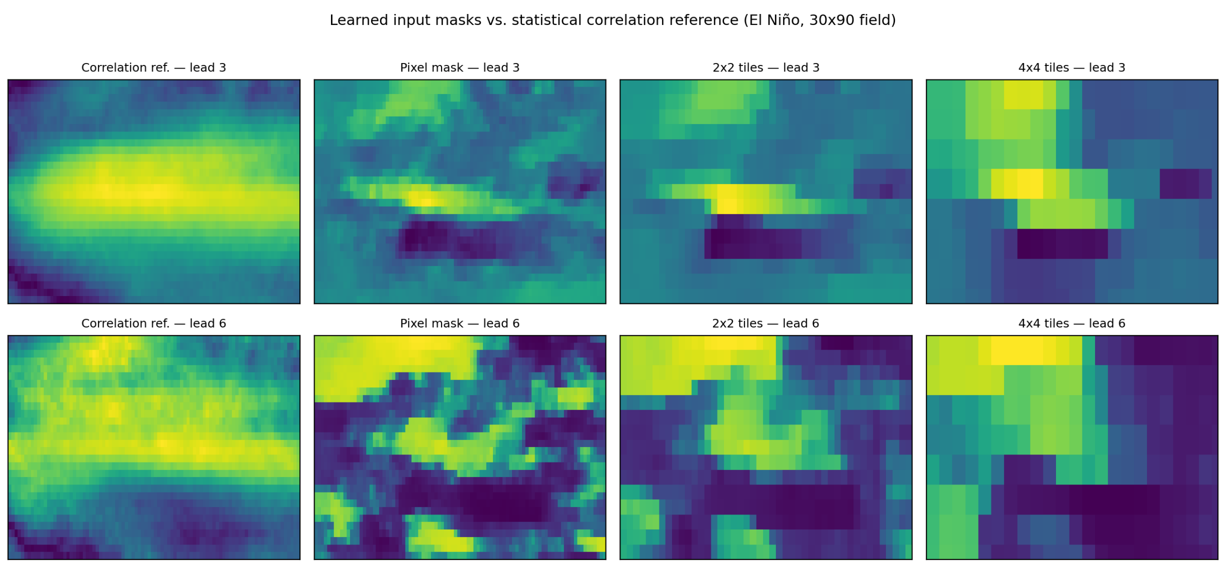

Figure 5: Learned input masks (pixel, 2×2, 4×4) beside the correlation reference at leads 3 and 6. At lead 6 the masks—especially the tiled ones—track the equatorial band; at lead 3 the redundancy of the signal leaves the attribution unconstrained.

4.5 Computational cost and the tiling knob

The cost of the TV penalty scales with the number of mask parameters, so on the native $0.25°$ field the pixel gate is expensive ($\approx 600\,\mathrm{s}$ per fit) while tiled gates run in seconds; on the 30×90 grid used above the backbone dominates and all gates train in $\approx 23\,\mathrm{s}$. Either way, tiling reduces the mask parameter count by a factor of K^2 and—as the lead-6 agreement shows—can yield *cleaner* attribution, providing a practical knob for trading spatial resolution against computational cost in operational forecasting.

5 Discussion

Our results support two claims and expose two honest limits. On the positive side, the controlled simulation shows that a total-variation smoothness prior *measurably and significantly improves the faithfulness* of learned in-model attribution, and does so not only for our sigmoid mask but, applied as a modular term, for L_0 and stochastic-gate baselines as well—total variation behaves as a general-purpose structured-sparsity prior for spatial attribution. Second, among learned gates our TV+sparsity mask occupies a favourable point on the faithfulness–accuracy frontier: it delivers faithful, spatially coherent attribution while preserving (indeed slightly improving) predictive performance, whereas sparsity-only gates buy sharper localization only by degrading the predictor they are meant to explain.

Two limitations temper these claims. First, the sparsity penalty rewards concentrating mask mass on the *cheapest* predictive region; when a non-causal feature is a cheap shortcut, the mask can be captured by it (Section 3.6). We therefore do not claim robustness to spurious correlation, and identify characterizing this failure—when does sparsity-driven masking resist versus succumb to shortcuts?—as important future work. Second, the El Niño study surfaces an *intrinsic* limit: when a spatially-dependent signal is highly redundant, many subregions are individually sufficient and the attribution is not uniquely determined, regardless of method or regularization. This is a property of the problem, not the estimator, but it bounds what any single attribution map can claim. Finally, we fixed the TV weight across methods in the modular study rather than tuning it per gate, our anomaly normalization uses the full-series climatology (a mild, common form of leakage), and the fused-lasso reference is collinearity-limited; each is a candidate for refinement.

Natural extensions include an uncertainty-aware variant that infers a posterior over the mask rather than a point estimate, connecting the structured-sparsity prior used here to the hierarchical Bayesian spatiotemporal tradition (Wikle & Zammit-Mangion 2023, Zammit-Mangion & Wikle 2020), and a temporal (three-dimensional) gate for spatiotemporal rather than purely spatial attribution.

6 Conclusion

We studied total-variation–regularized input masking as a route to faithful spatial attribution under spatial dependence. Framing the mask penalty as a structured-sparsity prior—linking total variation, the fused lasso, and Markov-random-field smoothness to a jointly-trained attribution gate—we showed on synthetic ground truth that the smoothness term significantly improves attribution faithfulness across learned gating mechanisms, and that our mask trades a negligible amount of localization for a large gain in predictive fidelity relative to sparsity-only gates. Applied to El Niño forecasting, the masks preserve predictive skill and align with a statistical reference where attribution is well-posed, while honestly exposing the limits of attribution when a spatially-dependent signal is highly redundant. We hope the structured-sparsity perspective on learned attribution proves useful for interpretable modeling of high-dimensional environmental fields.

7 Disclosure statement

The author declares no conflict of interest.

8 Data Availability Statement

Code and data available at: `https://github.com/drbob-richardson/cnn_sst`.

SUPPLEMENTARY MATERIAL

Code repository: Full implementation, training scripts, and experiment configs (GitHub ZIP).

Dataset files: Preprocessed versions of the datasets used in Section 3.

Additional figures: Ablation and sensitivity analysis figures not in the main paper.

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